BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. Page 53 of 94

Variables, Input, Output and One-Dimensional Arrays

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before turning to the answers.

1. What does bubble sort compare?

2. What does insertion sort build?

3. What is the first array index in OCR ERL?

4. Trace one bubble-sort pass through [3, 2, 1].

Today's objectives

Assign and update variables using OCR ERL. Identify input and output in short programs.

Key vocabulary - Read, cover, recall

variable
Named storage for a value in a program.

assignment
Storing or updating a variable's value.

one-dimensional array
Indexed collection of values stored under one name.

index
Number specifying an array element's position.

Write one definition in your own words:

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Learn and practise

New learning - Four key ideas

VARIABLE	A named value can change using assignment.
INPUT / OUTPUT	Read data from a user; display or send the result.
1D ARRAY	Related values stored at numbered indexes.
INDEXES	The first element has index 0, not index 1.

ARRAY ELEMENTS - INDEX AND VALUE ARE DIFFERENT

Index 0	Index 1	Index 2
Value 4	Value 7	Value 9

total = marks[0] + marks[1] = 4 + 7 = 11

I Do - Worked example

Trace the array and variable assignments in a short quiz-mark program.

1. Start: marks = [4, 7, 9]. Index 0 contains 4.
2. Read marks[0] + marks[1] = 4 + 7 = 11.
3. Change marks[0] to 5; indexes stay the same.
4. Recalculate: 5 + 7 = 12. Output 12.

We Do - Guided practice

Write the assignment to put 12 into score.

BLACK PEN - complete your explanation:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. Page 3 of 4

Think and write

Code First - Predict > Write > Test > Improve

Read the OCR Exam Reference Language code before testing it.

```
array fileMB = [2,5,8]
totalMB = fileMB[0] + fileMB[1]
print(totalMB)
```

1. Predict the total in MB.

2. Change fileMB[1] to 6; predict the new total.

3. Convert the revised 8 MB into KB (1 MB = 1000 KB).

You Do - Independent exam questions

Q1. State one difference between a variable and an array. [2 marks]

Q2. Given values = [7, 2, 5], state the values at indices 0 and 2. [2 marks]

Q3. Write OCR ERL assignments to set total to 0, add values[0] and values[1], then display total. [4 marks]

Exit Ticket - True or False?

Answer before checking Page 4.

1. The element at index 2 is normally the third element of an OCR ERL array. ☐ T ☐ F

2. Index 0 is the first array item. ☐ T ☐ F

3. Arrays and variables always store exactly the same number of values. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. Page 53 of 94

Review & improve

GREEN PEN: Mark fairly. Tick correct points and fix any mistakes.
Use the mark allocations; then rewrite one answer in full.

Do Now - Answers

Q1 Neighbouring values. ☐

Q2 A growing sorted section. ☐

Q3 Zero. ☐

Q4 [2, 3, 1] then [2, 1, 3]. ☐

Code First - Answers

1. Initial output: 7 MB (2 + 5). ☐

2. After change: 8 MB (2 + 6). ☐

3. 8 MB = 8,000 KB at 1000 KB per MB. ☐

We Do - Check the model

score = 12.

Exit Ticket - Check

1. TRUE 2. TRUE 3. FALSE

Explain one false statement using the key ideas.

You Do - Self-mark

Q1 Variable stores a value [1]; array stores indexed values [1]. ☐

Q2 Index 0 = 7 [1]; index 2 = 5 [1]. ☐

Q3 total=0 [1]; add values[0] [1]; add values[1] [1]; print(total) [1]. ☐

Try the question again

1. Copy the FULL question you got wrong (or found hardest):

2. In GREEN PEN, write a NEW answer. Include working if needed: